Steven Figler

Independent software engineer · Rust, ML inference, media systems

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I turn research code into software people can install and run. Since 2021 I have built, on my own and under the Step Up Gaming name, a native video engine that AI agents drive over MCP, Rust ports of open speech and TTS models, and the products and sites built on top of them, from first commit to deployment.

SKILLS

Languages	Rust, Python, TypeScript / JavaScript, HTML/CSS, SQL
ML & inference	Candle, ONNX Runtime, PyTorch (torch.fx), quantization (int8/Q8/FP16), WASM inference, ComfyUI, Qwen-Image
Systems & media	Skia, Vulkan, FFmpeg, audio DSP, parallel rendering, MCP servers, agent tooling
Web & infra	Next.js, React, Astro, FastAPI, Tauri, Cloudflare Workers / D1, GitHub Actions, Git

PROJECTS · STEP UP GAMING, 2021 – PRESENT

Director · native video engine that agents can drive

github.com/stepupgaming/director

- Compiles HTML/CSS and HyperFrames compositions into a typed, patchable scene document; a parallel Skia renderer paints it on a CPU pool or Vulkan and FFmpeg encodes. 102k lines of Rust across 10 crates.
- 2.13× faster 1080p export; supports 3D transforms with depth sorting, backdrop blur, masks, blend modes, text and audio mixing.
- Built an MCP server and revisioned host session: agents open, patch, inspect and export projects at about 11 ms per patch and 160 ms per 1080p frame on CPU.

Pocket TTS in Rust · first Rust port of Kyutai's Pocket TTS

128 stars · 29 forks

- Rewrote the Python model in Rust on Candle: one binary, streaming output (RTF 0.33), int8 weights, an OpenAI-compatible HTTP API, a React web UI, Python bindings and a WASM build. About 3× real time on CPU, 6× on Metal.

More model ports · PyTorch to Rust

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- **parakeet-candle:** NVIDIA Parakeet TDT 0.6B speech-to-text on Candle (v2 English, v3 multilingual) with token, word and sentence timestamps.
- **pycandle:** transpiler from PyTorch (via torch.fx) to Candle Rust; solves einsum, checks shapes, switches to flash attention.
- **ort-chatterbox:** Chatterbox Turbo TTS on ONNX Runtime with zero-shot voice cloning, FP32/FP16/Q8 and streaming. Also a private MiniMax H3 port.

ThumbForge · local-first AI thumbnail studio

desktop app

- Runs Qwen-Image-2.1 on the user's own ComfyUI: a brief becomes a storyboard, then drafts, then an automated feed check scoring contrast, focal point and phone-size legibility. Python, FastAPI, Next.js.

ClipFactory · local batch clipper for vertical video

stepupgaming.github.io/clipfactory

- Transcript-first clip selection, multi-beat FFmpeg rendering, deterministic QA gates and style transfer from a reference video; 482 tests. Public demo runs fully in the browser.

NoFaceTube & StepUpAI · production web products

nofacetube.com · stepupai.com

- NoFaceTube: editorial reference site with a public quality rubric and dated platform-rules tracker (Astro 5, React islands, Cloudflare Worker + D1).
- StepUpAI: site, content pipeline and internal tools for the studio, run without a team (Next.js, React, TypeScript, Cloudflare).

ALSO BUILT

- **grits**, a Git-native issue tracker for AI agents · **comfy-workflows**, typed code-first ComfyUI workflows · **magenta-windows**, a Windows player for Magenta RealTime · **VNCCS-Desktop** (Tauri, Next.js) · **tauri-rustdocs-candle**, semantic rustdoc search

EDUCATION

Self-directed: Rust systems programming, ML inference and model porting (2021 – present) · High school diploma